Introduction to Computer Networking

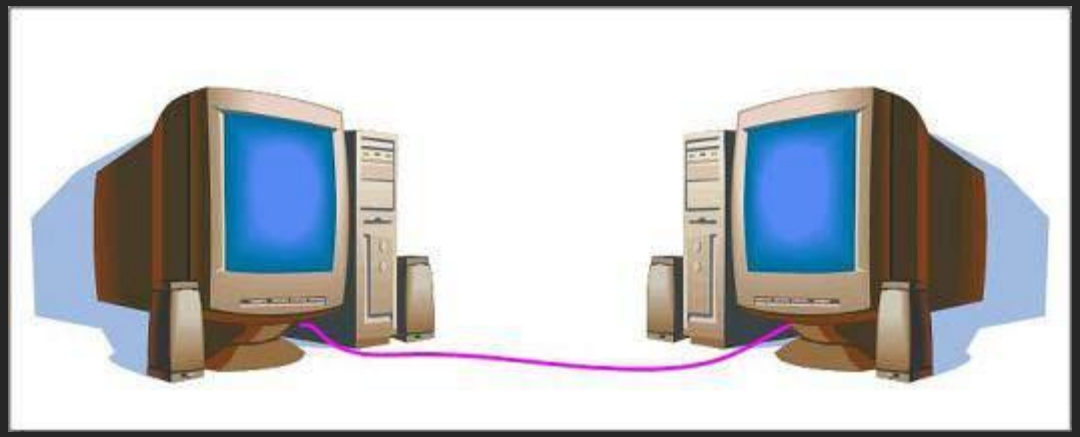
Introduction to Computer Networking

A *network* is simply two or more computers that are linked together.

NETWORK - DEFINITION

- A network can be defined as two or more computers connected together using a connection medium in such a way that they can communicate and share resources.

- The purpose of a network is to share resources.



RESOURCES?

A resource may be:

- A file
- A folder
- A printer
- A disk drive
- Or just about anything else that exists on a computer.

What is Networking?

Networking is the term that describes the processes involved in designing, implementing, upgrading, managing and working with networks and network technologies.

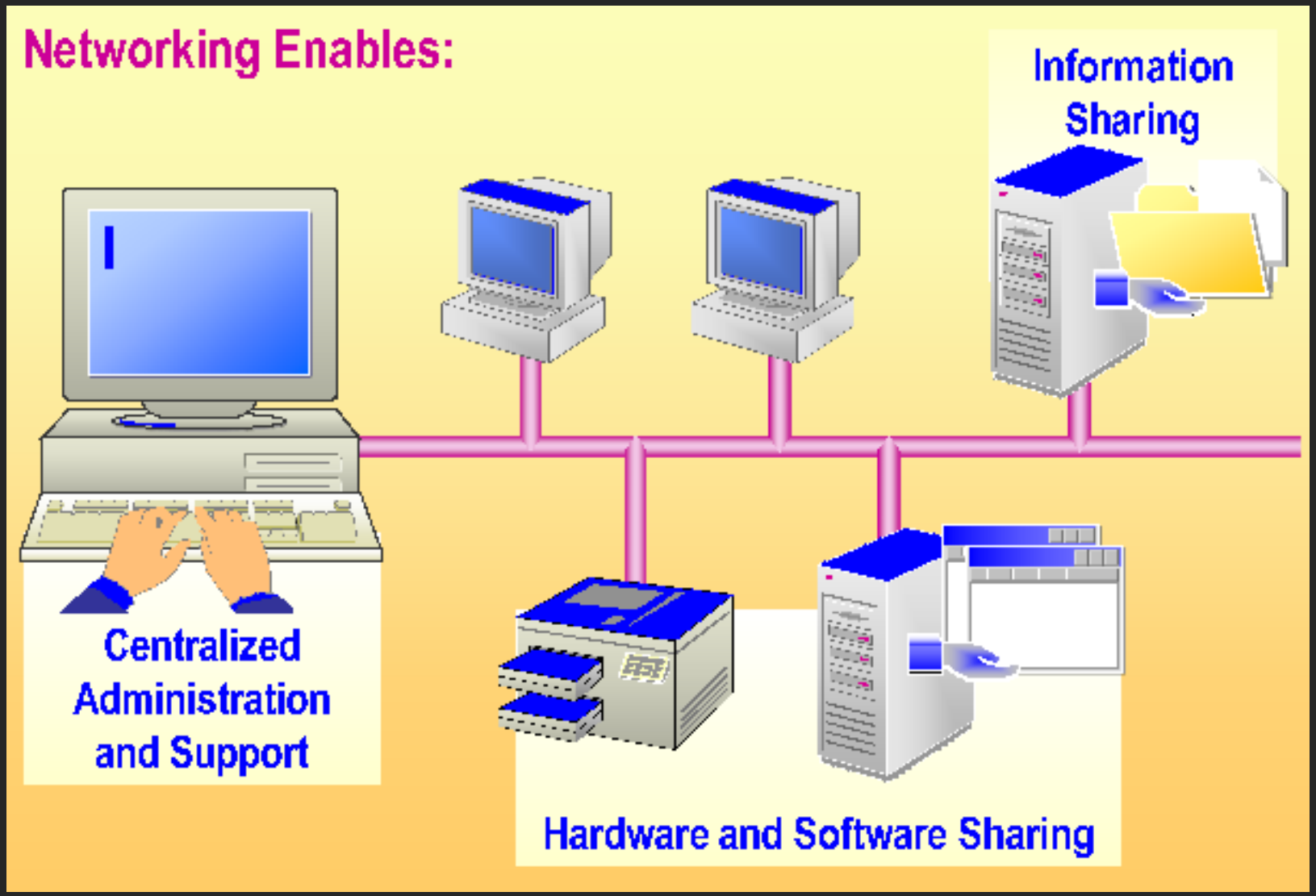
BENEFITS OF NETWORKING

- *Information sharing:* Authorized users can use other computers on the network to access and share information and data. This could include special group projects, databases, etc.
- *Hardware sharing:* One device connected to a network, such as a printer or scanner, can be shared by many users.
- *Software sharing:* Instead of purchasing and installing a software program on each computer, it can be installed on the server. All of the users can then access the program from a single location.
- *Collaborative environment:* Users can work together on group projects by combining the power and capabilities of diverse equipment.

BENEFITS OF NETWORKING

- *Internet Access & Sharing:* Browse World Wide Web (WWW) to access recourses and share information.
- *Entertainment :* Online movies, music, games, Facebook, etc.

BENEFITS OF NETWORKING



COST & SECURITY OF NETWORK

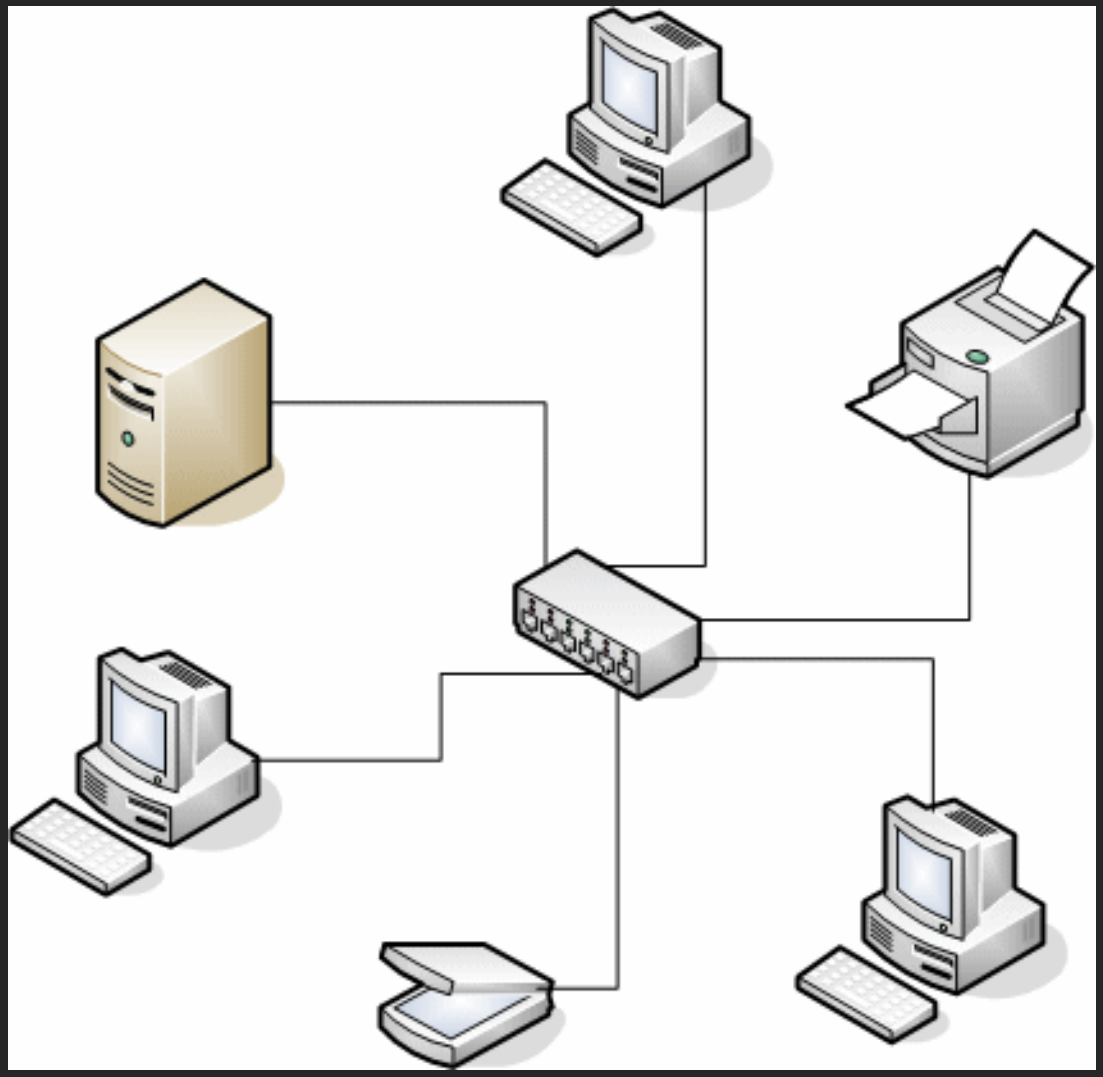
- Network Hardware, Software and Setup Costs
- The security of a computer network is challenged everyday by:
  - Virus/worms attacks
  - Computer hackers
  - Equipment malfunctions (HW Malfunction, SW Crash, DoS)

TYPES OF NETWORKS

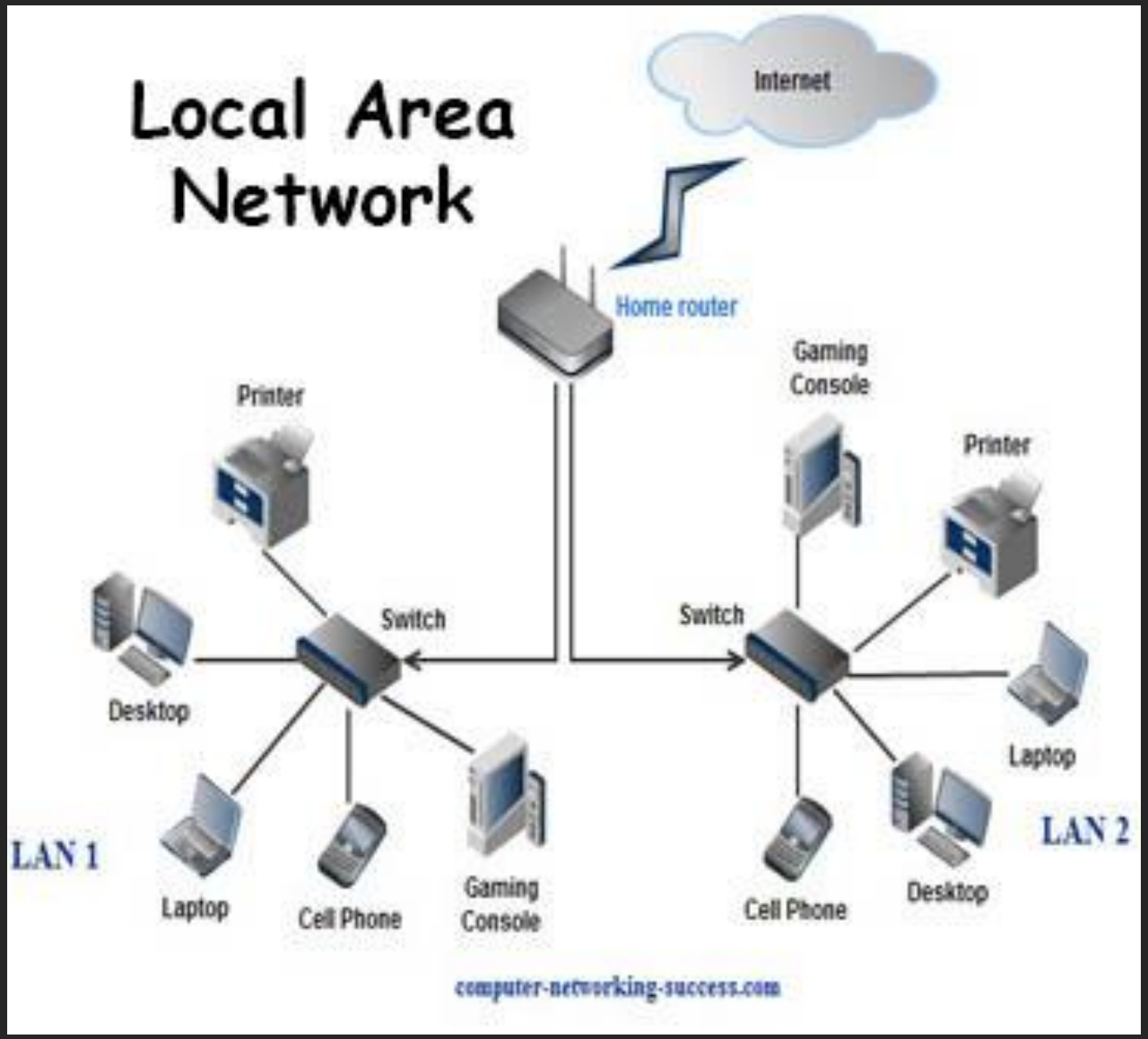
Local Area Networks (LANs):

- A **local area network (LAN)** is a computer network covering a small geographic area, like a home, office, or group of buildings.

NETWORK TO SHARE RESOURCES



Local Area Networks (LANs)

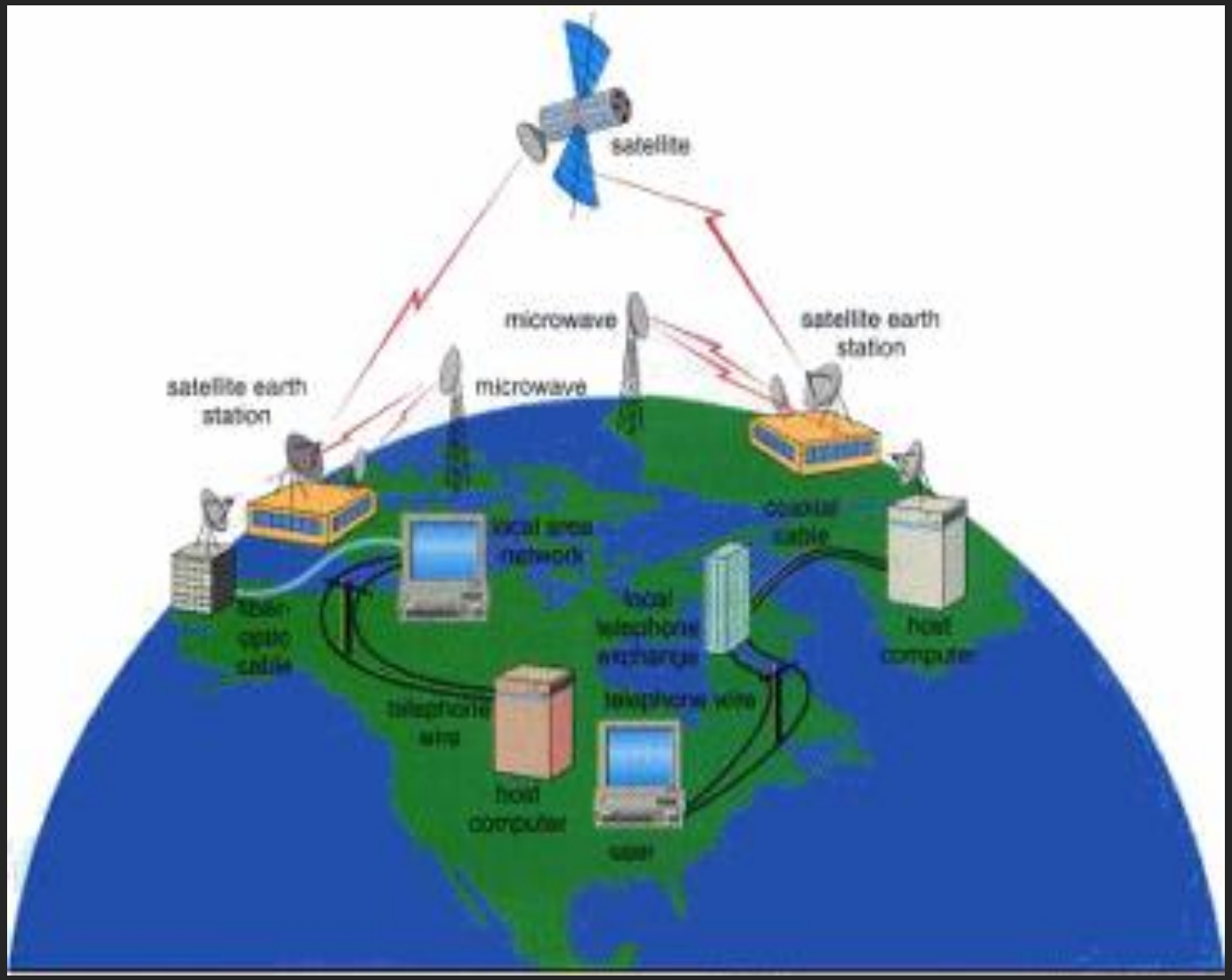


TYPES OF NETWORKS

Wide Area Networks (WANs):

- **Wide Area Network** (**WAN**) is a computer network that covers a broad area.
- WANs are used to connect LANs and other types of networks together.
- The largest and most well-known example of a WAN is the Internet.

WIDE AREA NETWORK - WAN

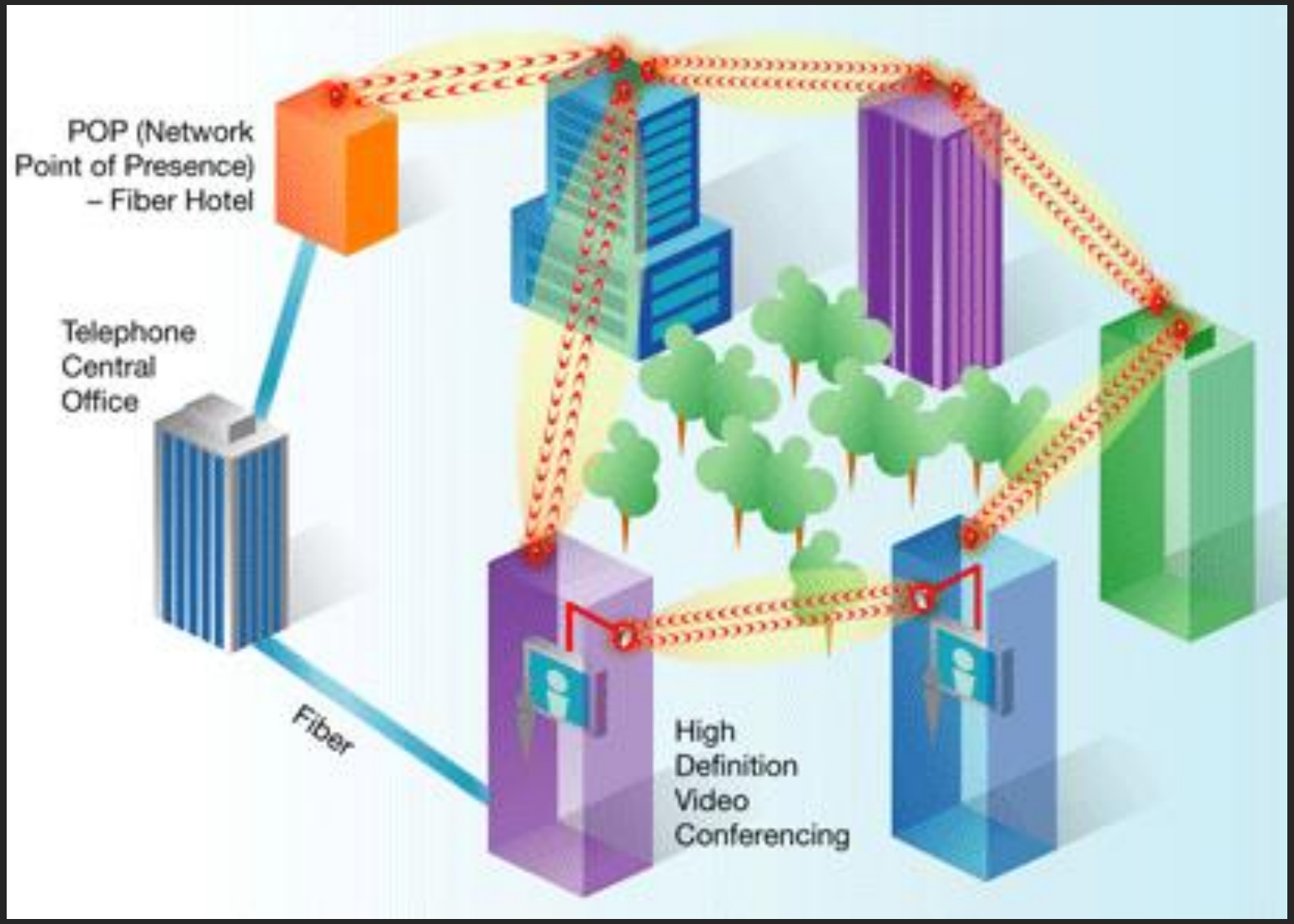


TYPES OF NETWORKS

Metropolitan Area Network (MAN):

- A metropolitan area network (MAN) is a network that interconnects users with computer resources in a geographic area or a region larger than local area network (LAN) but smaller than a wide area network (WAN).

METROPOLITAN AREA NETWORKS



MORE TYPES OF NETWORKS

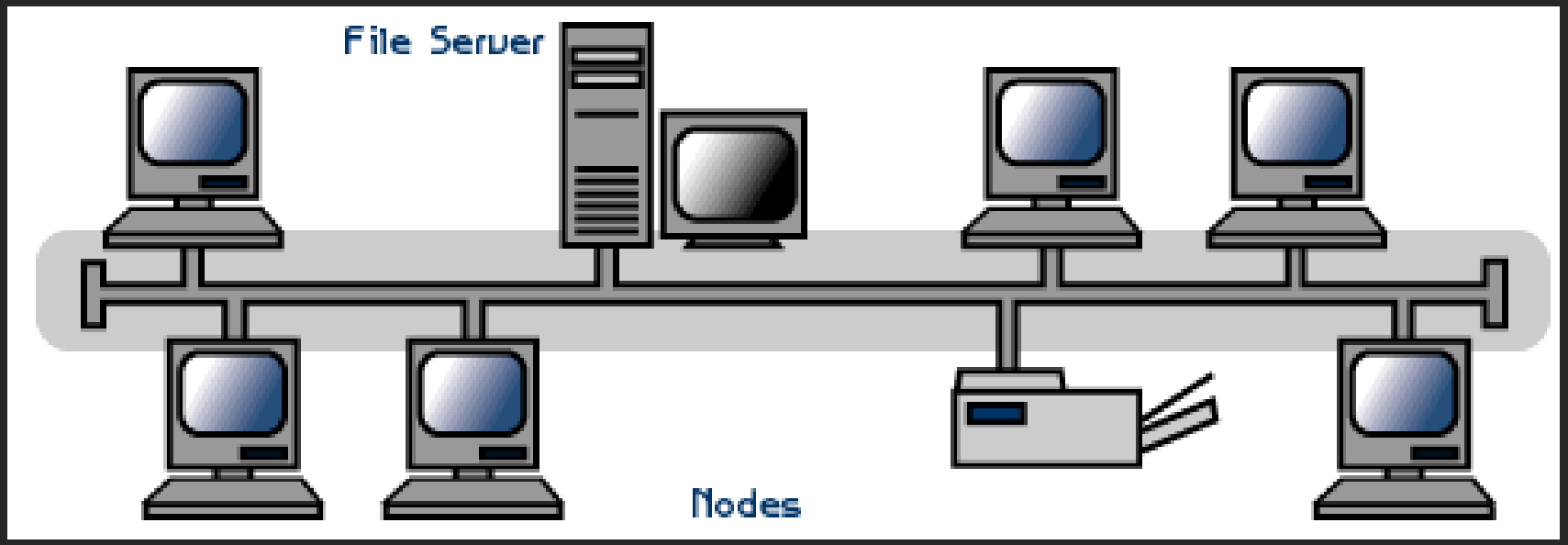
- Campus Area Network – CAN
- Personal Area Network – PAN
- Storage Area Network - SAN

NETWORK TOPOLOGIES

Networks can be mainly broken into 4 topologies.

- BUS TOPOLOGY
- RING TOPOLOGY
- STAR TOPOLOGY
- MESH TOPOLOGY

BUS TOPOLOGY



Bus topology allows information to be directed from one computer to the other. Lots of binary collision though.

RING TOPOLOGY

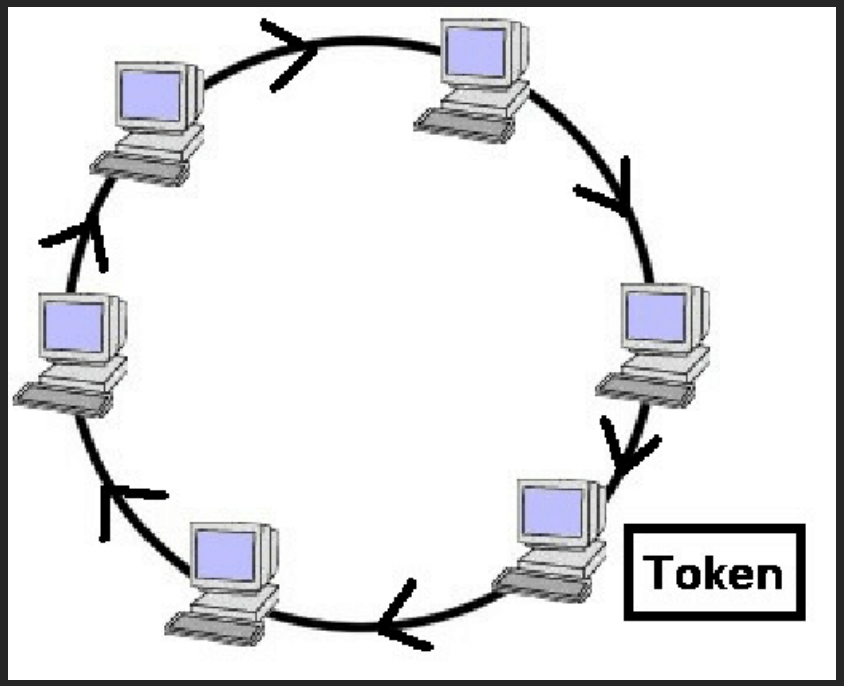
Ring topology- uses a token to pass information from one computer to the other.

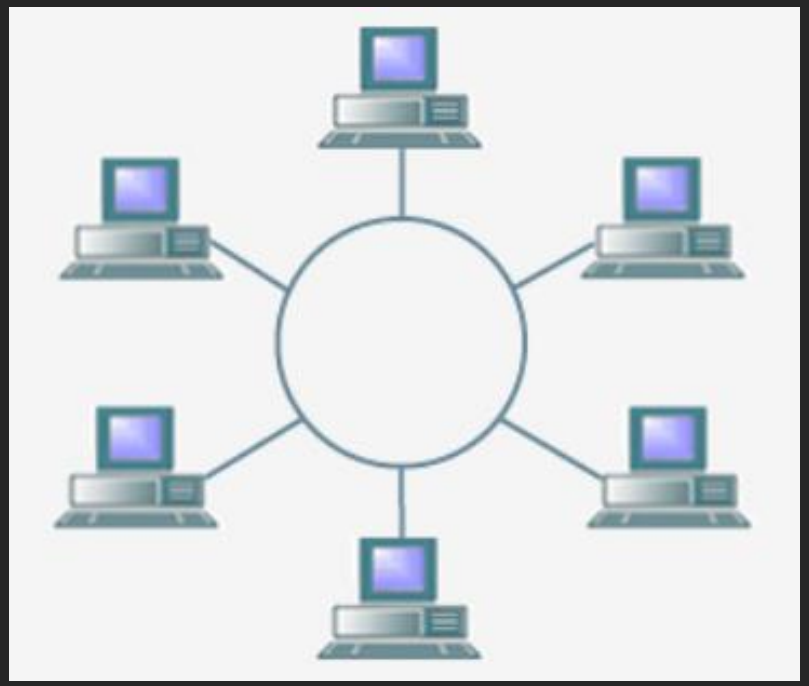
A token is attached to the message by the Sender to identify which computer should Receive the message.

As the message moves Around the ring, each computer examines the token.

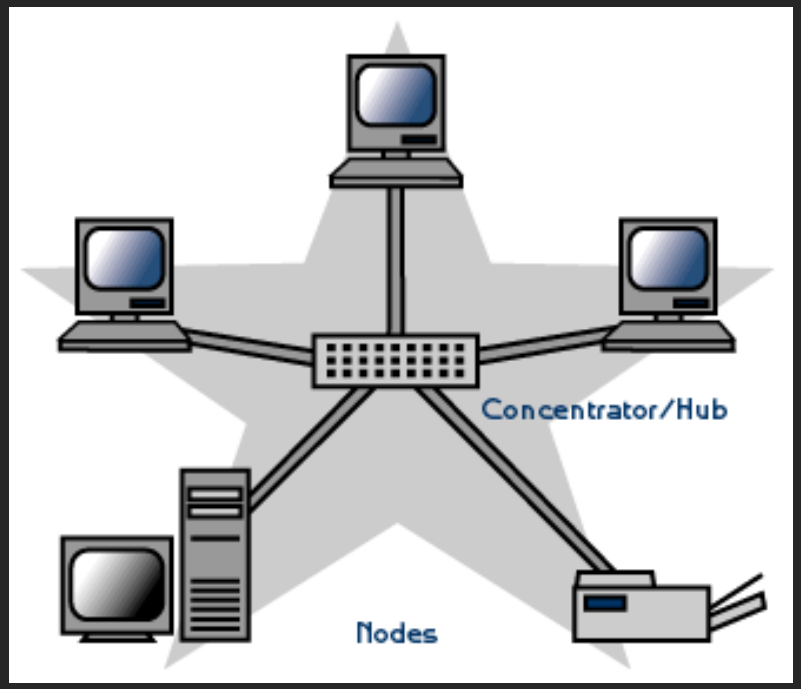
If the computer identifies the token as its own, then it will process the information.

RING TOPOLOGY





STAR TOPOLOGY

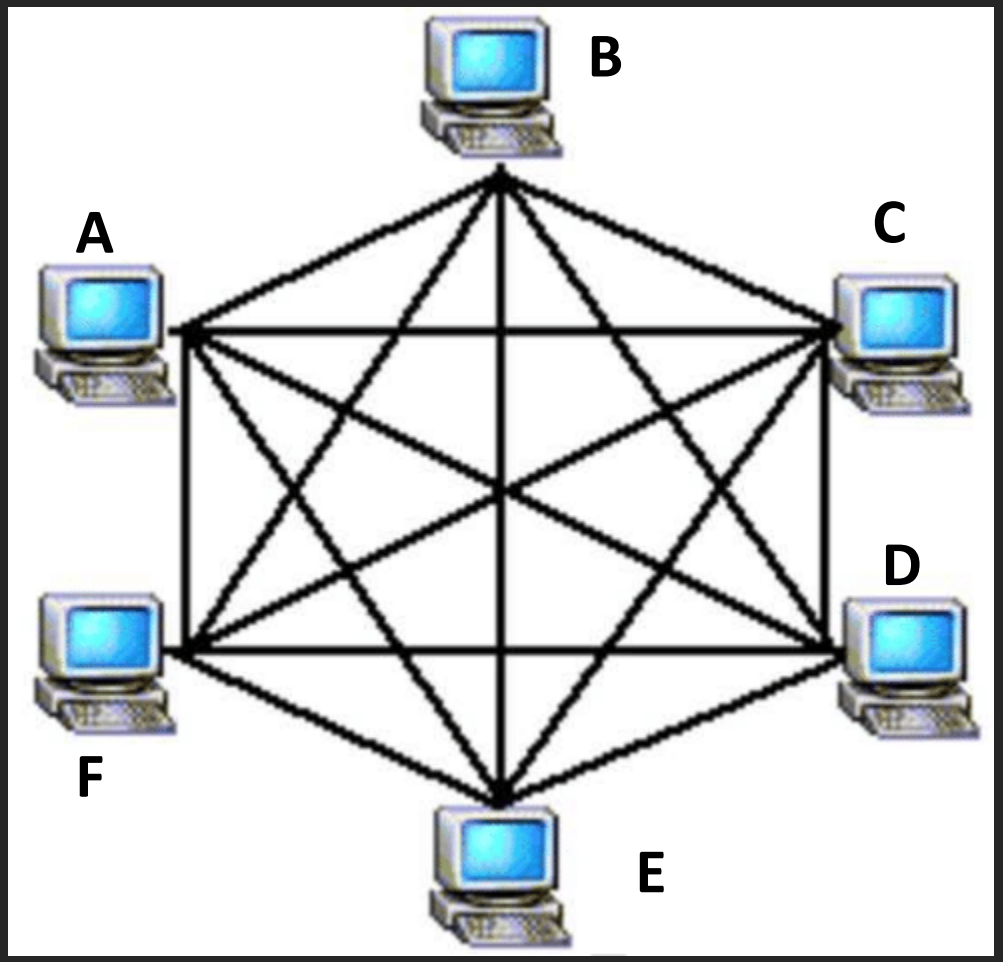


Star topology is the most common type Used. All computers are attached to a switch. Less collisions and most efficient.

MESH TOPOLOGY

Full Mesh Topology

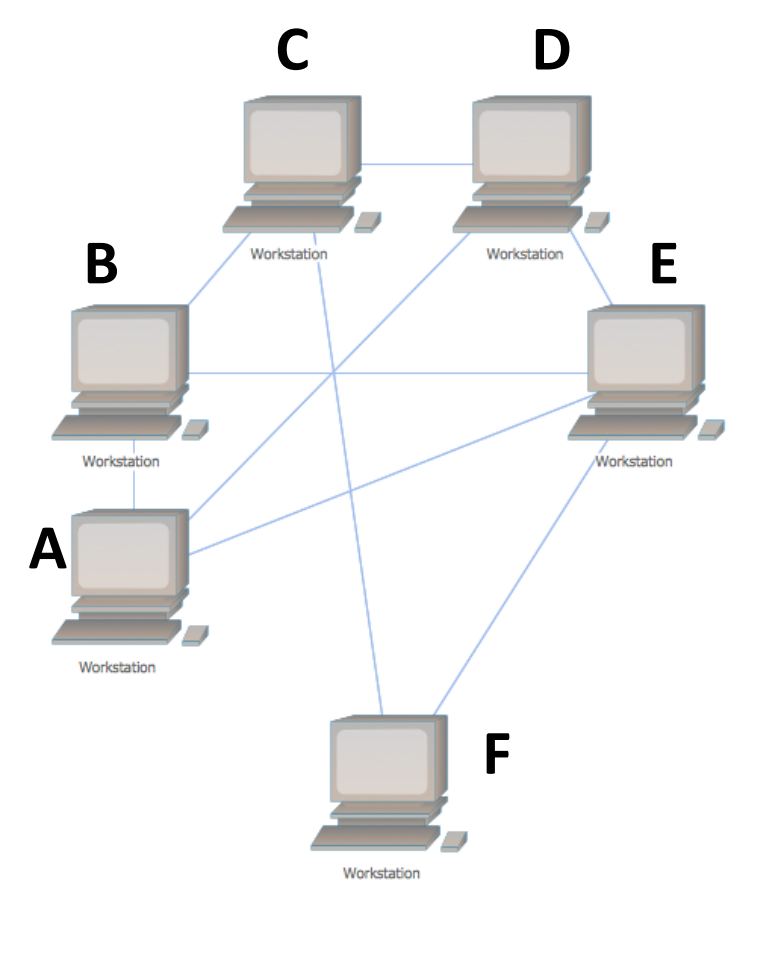
All the computers in the network have connections with all the other computers.



MESH TOPOLOGY

Half Mesh Topology

Computers have multiple connections, but not with all the computers.



NETWORK/COMMUNICATION PROTOCOLS

- ***A protocol*** is simply an agreed-on set of rules and procedures for transmitting data between two or more devices.
  - Eg: HTTP, SMTP, FTP, Telnet, ssh, ICMP

OSI 7-Layer Model

- In the 70's, computer networks were ad hoc, vendor-specific.
- OSI Model is a conceptual model to describes how a computer system communicates over a network.

Open Systems Interconnection model

- Divides the tasks involved in moving information between networked computers into 7 task groups
- Each task group is assigned a layer

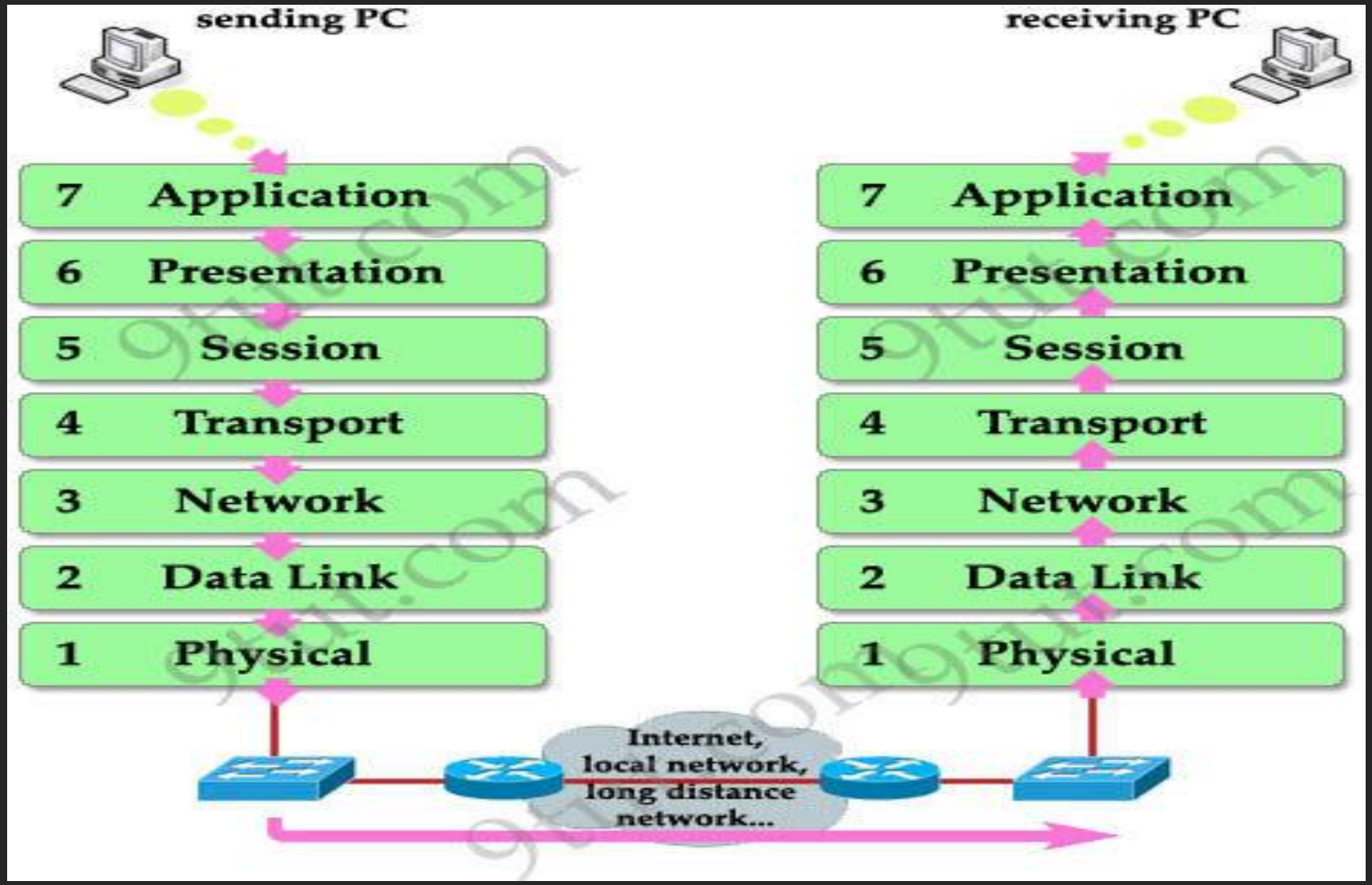
Each layer is reasonably self-contained, so

- Can be implemented independently
- Changes/updates to a layer need not effect other layers

OSI MODEL

The seven layers of the **Open Systems Interconnection** (OSI) Model are:

OSI MODEL


sending PC
receiving PC
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical
Internet, local network, long distance network...

OSI 7-Layer Model

- Application layer
  - describes how applications will communicate
    e.g., HTTP, FTP, Telnet, SMTP
- Presentation layer
  - describes the form of data being transferred & ensures that it will be readable by receiver
    e.g., data representation, data compression, encryption
- Session layer
  - Manages communication sessions
    e.g., coordinates requests/responses (“traffic flow”)

OSI 7-Layer Model

- Transport layer
  - describes the quality and nature of data delivery

    e.g. message segmentation, message acknowledgment, how retransmissions are used to ensure delivery
- Network layer
  - Selecting the best link that can deliver data across a network

    e.g. Routing, IP addressing

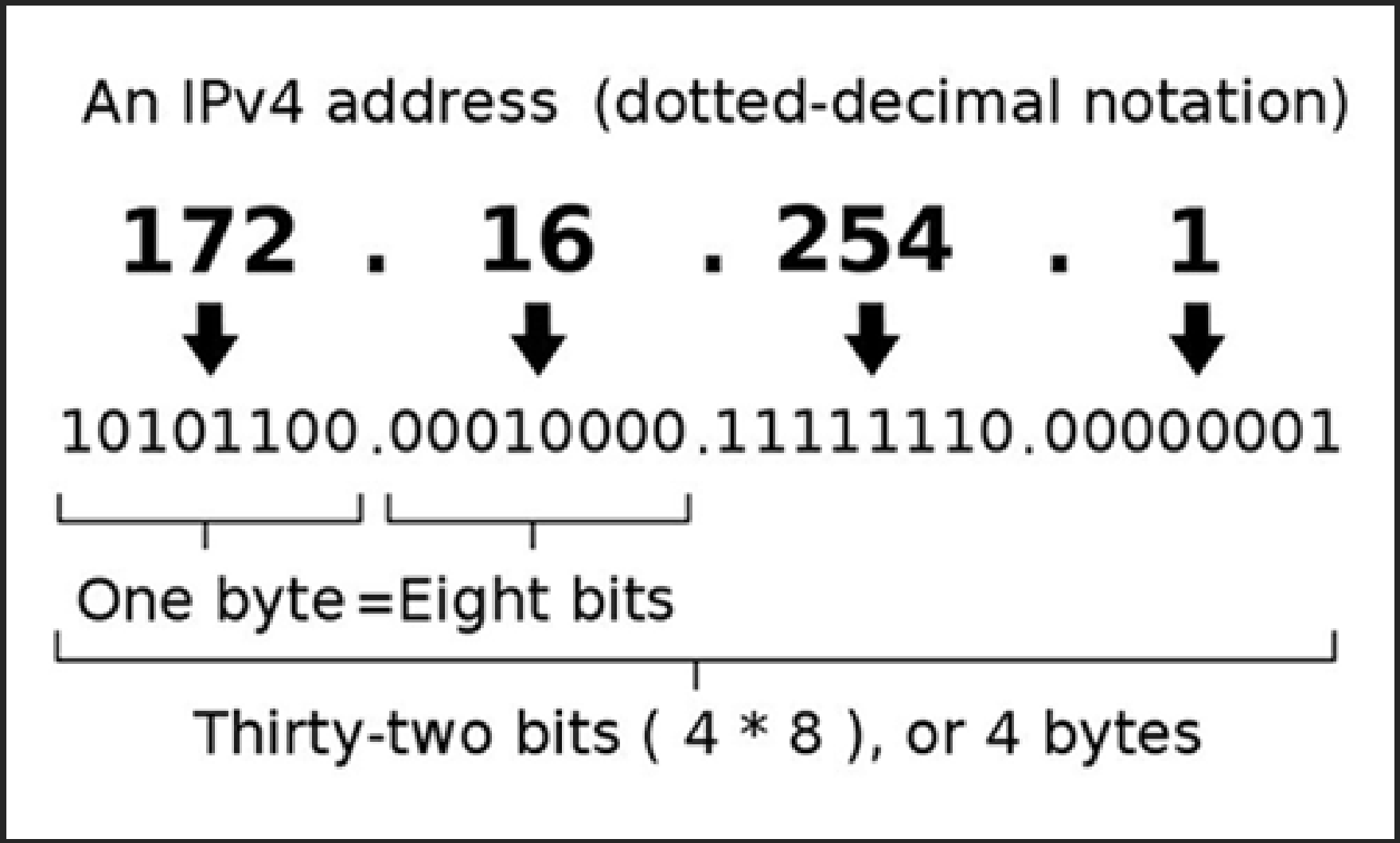
OSI 7-Layer Model

- Data Link layer
  - Data delivery within a local network
    e.g. Switching, MAC address.
- Physical layer
  - describes the physical & electrical properties of the communications media
    e.g. voltage levels, data rates, max distances

IP ADDRESS

- An IP address is a 32-bit address.
- The IP addresses are used to uniquely identify a host
- IP stands for Internet Protocol and is basically an address that the packets will be sent to.
- An example would be 192.168.1.10

IP ADDRESS


An IPv4 address (dotted-decimal notation)
172 . 16 . 254 . 1
10101100.00010000.11111110.00000001
One byte =Eight bits
Thirty-two bits (4 * 8), or 4 bytes

IP ADDRESS

If you look at the IP Address, the number are broken into different categories.

10. 27.61.137

Network

Hosts

IP ADDRESS CLASSES

Address Class	RANGE	Default Subnet Mask
A	1.0.0.0 to 126.255.255.255	255.0.0.0
B	128.0.0.0 to 191.255.255.255	255.255.0.0
C	192.0.0.0 to 223.255.255.255	255.255.255.0

SUBNET MASK

A Subnet mask is a 32-bit number that masks an IP address, and divides the IP address into network address and host address.

Class A – 255.0.0.0

Class B – 255.255.0.0

Class C – 255.255.255.0

IP ADDRESS

What class are they?

192.168.10.10

10.10.10.5

123.100.78.72

203.189.78.18

172.17.50.50

IP ADDRESS

Question

Convert the following IP to binary:

192.50.10.17

Convert the following IP to Decimal:

10000001. 0010001.00011001.00001001

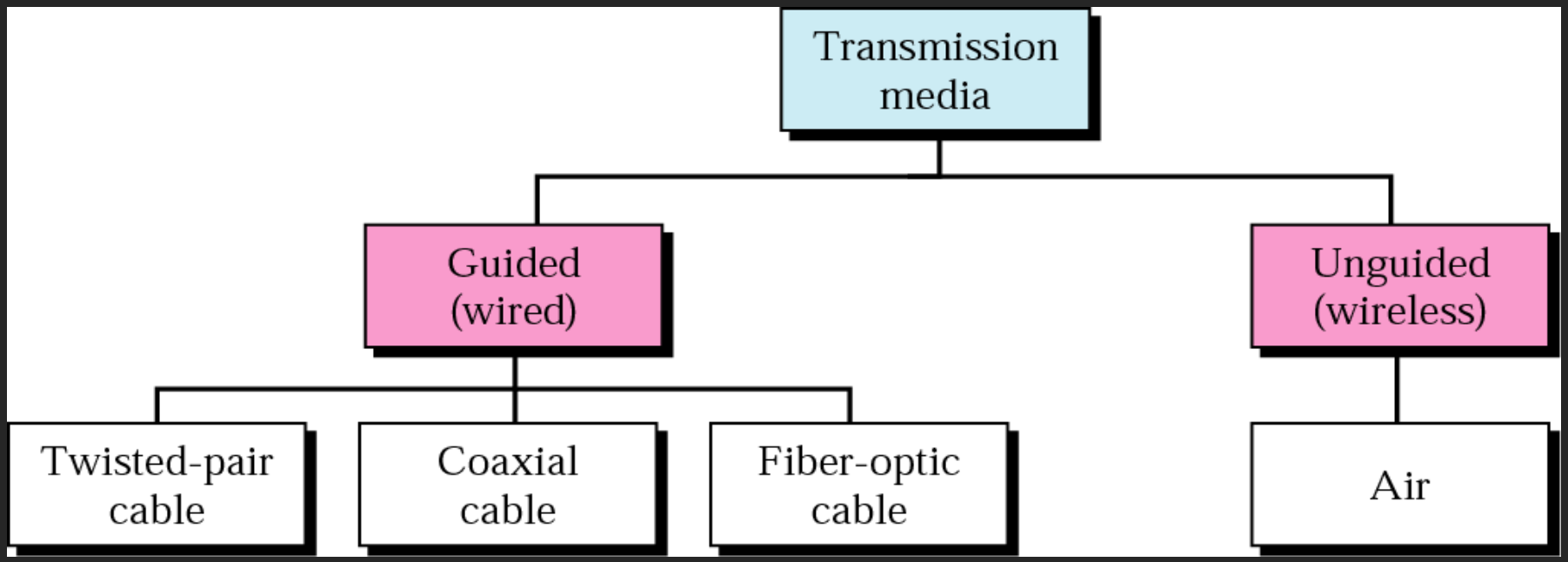
IPv6

- **Internet Protocol version 6** (**IPv6**) is the most recent version of the Internet Protocol (IP)
- IPv6 extends address size to 128 bits
- IPv6 was developed by the Internet Engineering Task Force (IETF) to overcome the limitations of IPv4.
- Eg: 2001:0db8:85a3:0042:1000:8a2e:0370:7334

TRANSMISSION MEDIA

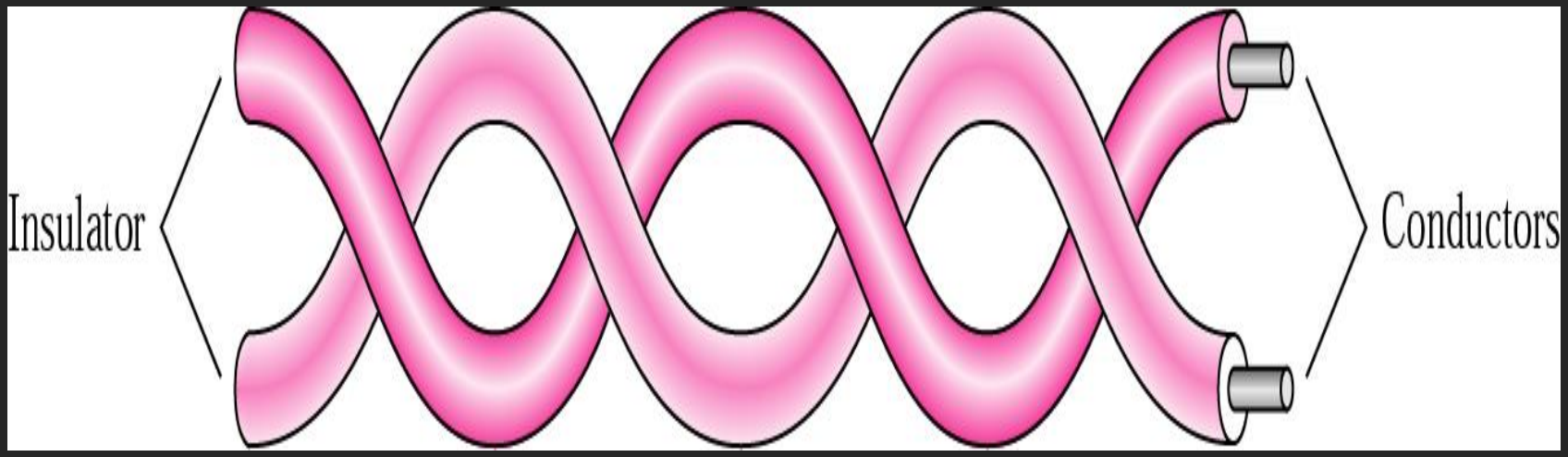
- The transmission medium is the path by which a message travels from sender to receiver.
- Computers and telecommunication devices use signals to represent data.
- These signals are transmitted from a device to another in the form of electromagnetic energy.
- Examples : radio waves, infrared light, visible light..

CLASSES OF TRANSMISSION MEDIA



TWISTED-PAIR CABLE

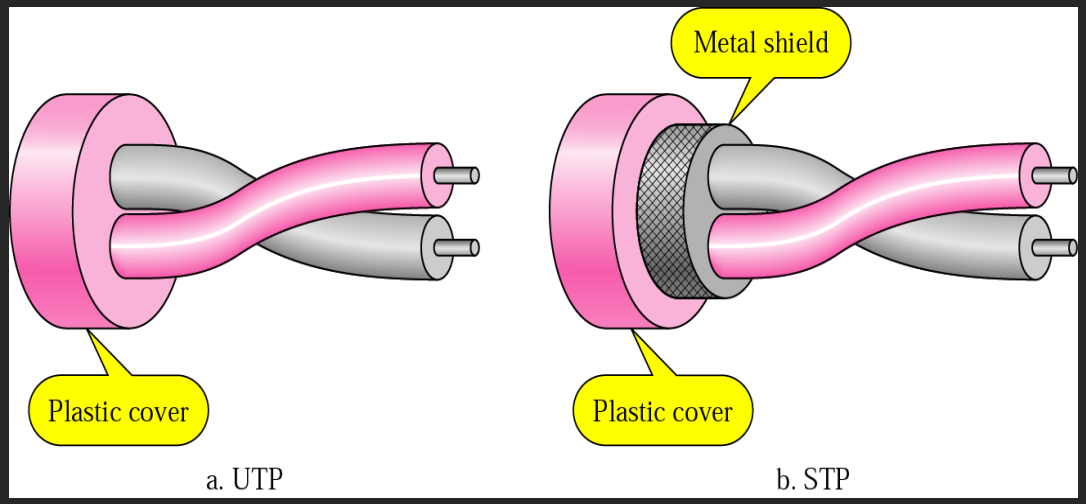
- Twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.
- Twisted-pair cable comes in two forms: **Unshielded** and **Shielded**
- The twisting helps to reduce the interference (noise) and crosstalk.

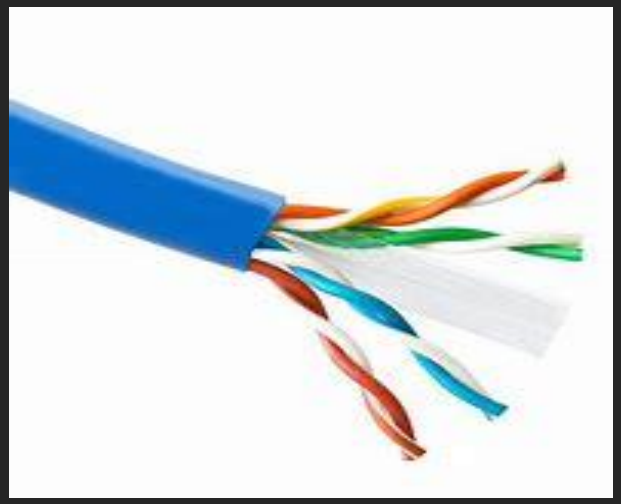


TWISTED-PAIR CABLE

Shielded Twisted Pair (STP) – Used In Phone Lines, Networks

Unshielded Twisted Pair (UTP) - Used In Phone Lines, Networks





CAT CABLES

UTP (Unshielded Twisted Pair) type of cable used for Ethernet wiring are Cat3, Cat4, Cat5, Cat5e, Cat6, and Cat7.

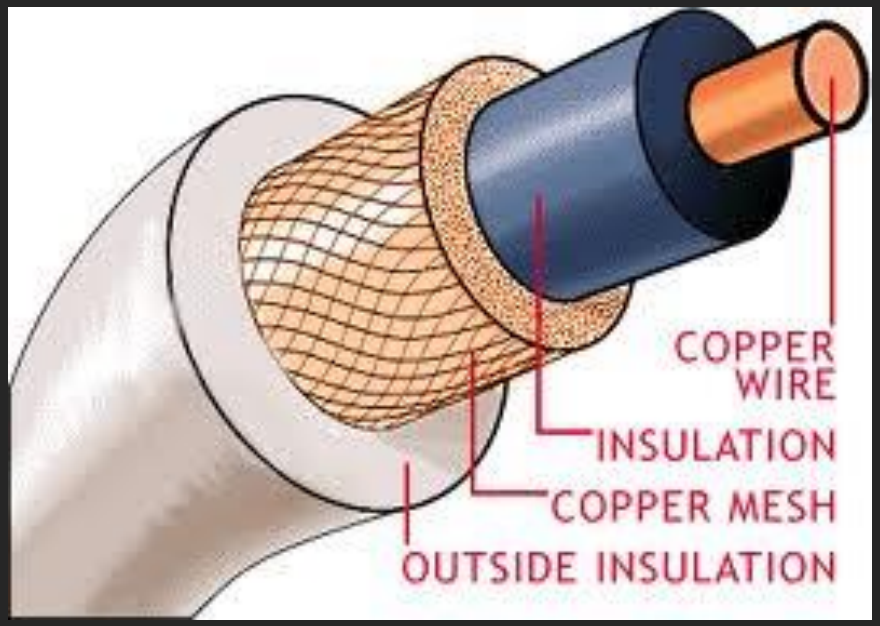
CAT CABLES

Category	Bandwidth	Data Rate	Digital/Analog	Use
1	very low	< 100 kbps	Analog	Telephone
2	< 2 MHz	2 Mbps	Analog/digital	T-1 lines
3	16 MHz	10 Mbps	Digital	LANs
4	20 MHz	20 Mbps	Digital	LANs
5	100 MHz	100 Mbps	Digital	LANs
6 (draft)	200 MHz	200 Mbps	Digital	LANs
7 (draft)	600 MHz	600 Mbps	Digital	LANs

COAXIAL CABLES

- Coaxial cable carries signals of higher frequency ranges than twisted-pair cable.

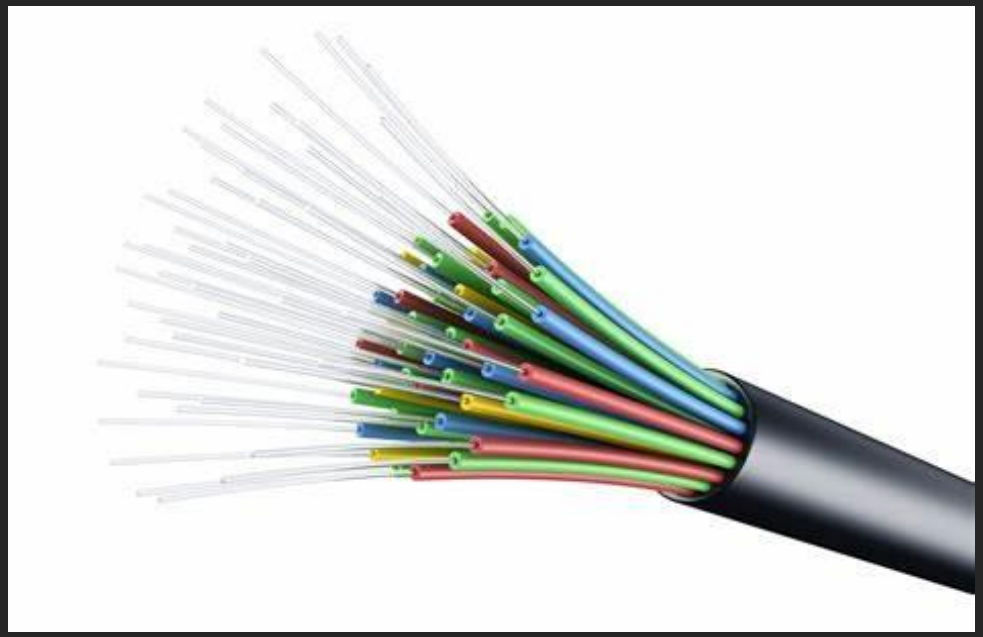
- Coaxial Cable standards: RG-8, RG-9, RG-11 are used in thick Ethernet RG-58 Used in thin Ethernet RG-59 Used for TV



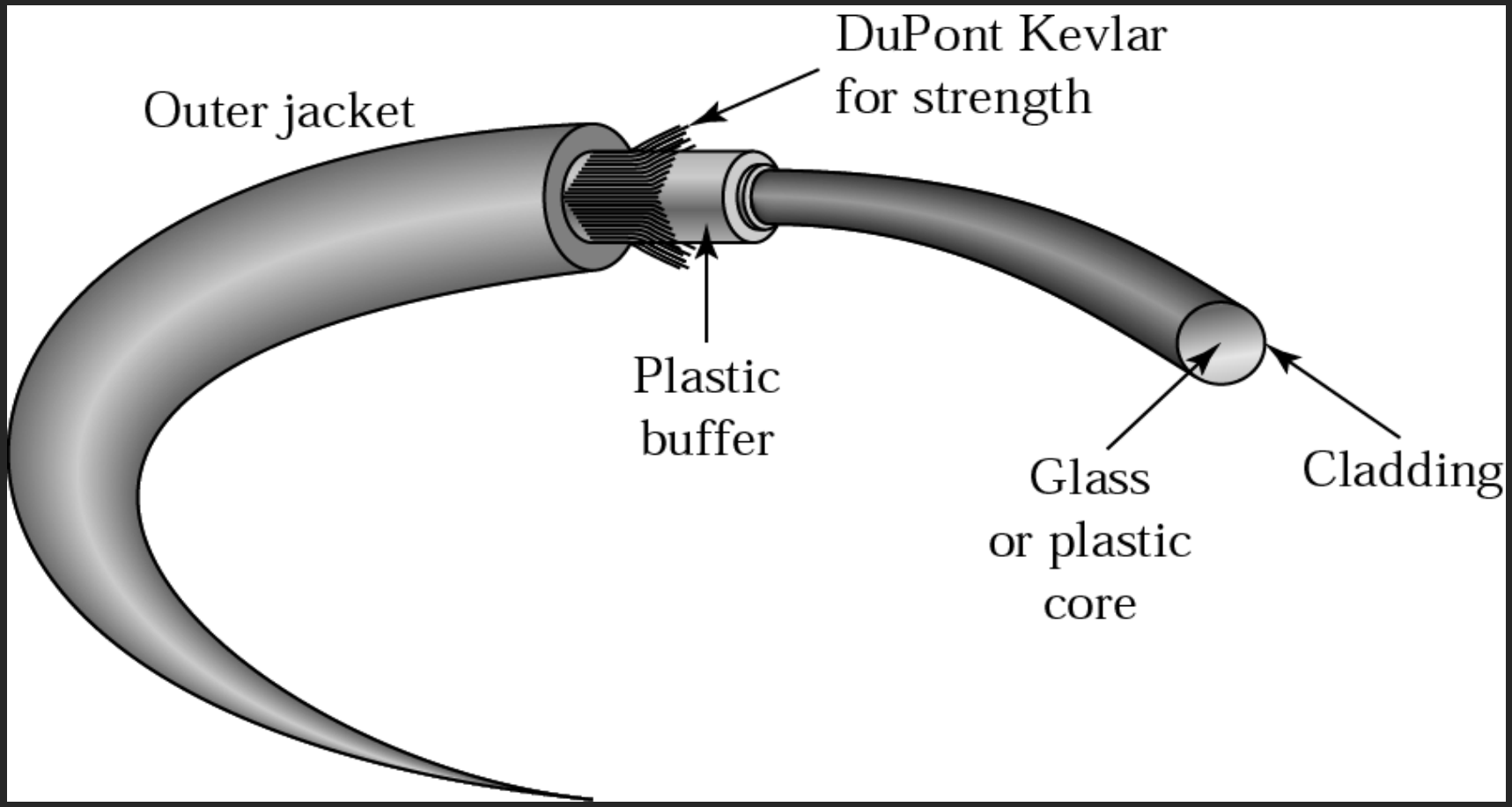
OPTICAL FIBER

- Optical fiber is made of glass or plastic and transmits signals in the form of **light**.

- Light, a form of electromagnetic energy, travels at 300,000 Kilometers/second in a vacuum.



FIBER CONSTRUCTION



Network Peripherals

Network Devices

- Network Interface Card (NIC) (LAN card)
  - Usually 100 MB/s
- Wireless LAN card
  - Usually 11 MB/s

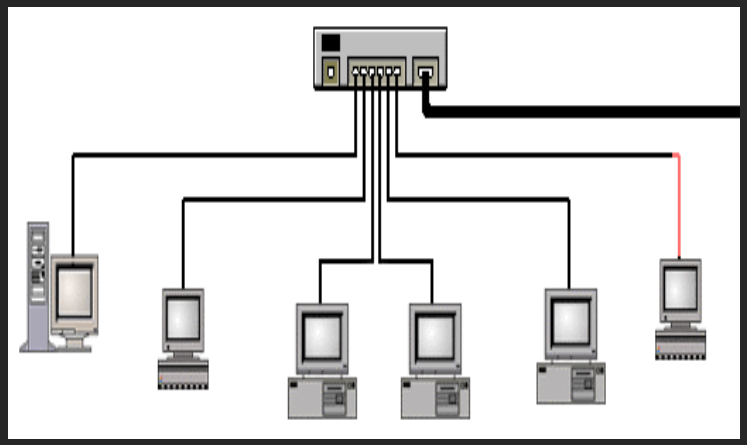
Network Devices

- Connectors
  - RJ-45
    - For connecting twisted pairs LAN cable
  - RJ-11
    - For connecting telephone lines
  - BNC connector
    - For connecting coaxial cables

HUBS

Hubs interconnects PCs together.

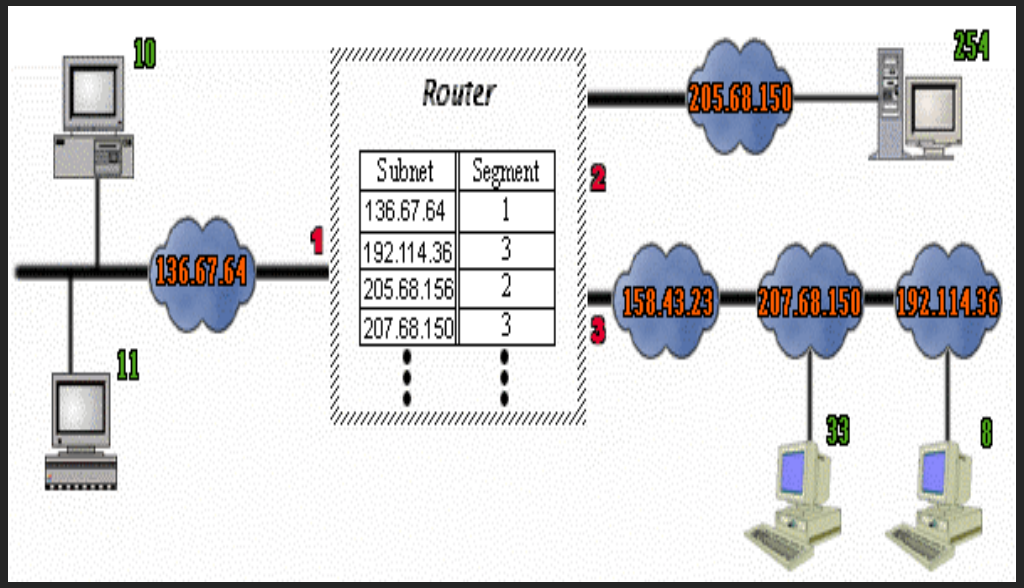
Hub's major function is to replicate data it receives from one device attached to it to all others (Broadcast system)



SWITCHES

A **network switch** is a small hardware device that interconnects multiple computers together within one local area network (LAN). Technically, network switches operate at layer two (Data Link Layer) of the OSI model.

ROUTERS



- Routers connect two or more networks and forward data packets between them. When data arrives from one of the segments, the router decides, according to it's routing table, to which segment to forward that data.

Troubleshooting

- Check Physical Network Connection
- Check whether your PC has an IP address
- You can check pinging an end device
- Make sure your firewall has the right permission (Ex. icmp is allowed to accept ping, RDP is allowed for remote desktop)
- You can telnet to a service (Ex. telnet www.nibm.lk 80)
- Sometimes simply rebooting your PC/Router may fix the issue.

Troubleshooting

Select Command Prompt

```
C:\>
C:\>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : nibm.local
   Link-local IPv6 Address . . . . . : fe80::6c00:8e92:9cb8:5b4%11
   IPv4 Address. . . . . . . . . . . : 192.168.12.127
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.12.254

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6922:67ea:c8cb:18e6%21
   IPv4 Address. . . . . . . . . . . : 10.0.3.20
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Troubleshooting

Command Prompt

```
C:\>
C:\>ping 192.168.16.34

Pinging 192.168.16.34 with 32 bytes of data:
Reply from 192.168.16.34: bytes=32 time<1ms TTL=63
Reply from 192.168.16.34: bytes=32 time<1ms TTL=63
Reply from 192.168.16.34: bytes=32 time<1ms TTL=63
Reply from 192.168.16.34: bytes=32 time<1ms TTL=63

Ping statistics for 192.168.16.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>ping 192.168.16.30

Pinging 192.168.16.30 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.16.30:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

A Basic Network:

- Have two or more PCs
- A switch
- Cables to connect (T/P - UTP cables)
- Assign PCs a IP address (In the same IP class and range)

- End -